Back Exercises & Answers

NCERT Science – Class X (CBSE)

A cumulative set of end-of-chapter questions from **Chapters 2, 3, 9 and 10**, with concise, exam-focused answers in CBSE format.

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Answers are kept short and to the point — ideal for quick revision and board-exam practice. Ray diagrams (Ch 9–10) should be drawn by the student as described.

CHAPTER 2 Acids, Bases and Salts

Section A — Multiple Choice Questions

Choose the correct option for the following questions.

Q1. A solution turns red litmus blue; its pH is likely to be

(a) 1
(b) 4
(c) 5
(d) 10 **[Correct]**

Ans. Bases turn red litmus blue and have pH > 7; among the options only pH 10 is basic.

Q2. A solution reacts with crushed egg-shells to give a gas that turns lime-water milky. The solution contains

(a) NaCl
(b) HCl **[Correct]**
(c) LiCl
(d) KCl

Ans. Egg-shells are $CaCO_3$; HCl reacts with it to release CO_2, which turns lime-water milky.

Q3. 10 mL of NaOH is completely neutralised by 8 mL of a given HCl solution. If 20 mL of the same NaOH is taken, the HCl required will be

(a) 4 mL
(b) 8 mL
(c) 12 mL
(d) 16 mL **[Correct]**

Ans. NaOH doubles, so HCl also doubles → 16 mL.

Q4. Which type of medicine is used for treating indigestion?

(a) Antibiotic
(b) Analgesic
(c) Antacid **[Correct]**
(d) Antiseptic

Ans. Indigestion is due to excess acid; antacids (mild bases) neutralise it.

Section B — Short / Numerical Questions

Q5. Write word equations and then balanced equations for the reaction when –
(a) dilute sulphuric acid reacts with zinc granules.
(b) dilute hydrochloric acid reacts with magnesium ribbon.
(c) dilute sulphuric acid reacts with aluminium powder.
(d) dilute hydrochloric acid reacts with iron filings.

Ans. (a) Zinc + Sulphuric acid → Zinc sulphate + Hydrogen
$Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2\uparrow$
(b) Magnesium + Hydrochloric acid → Magnesium chloride + Hydrogen
$Mg + 2HCl \rightarrow MgCl_2 + H_2\uparrow$
(c) Aluminium + Sulphuric acid → Aluminium sulphate + Hydrogen
$2Al + 3H_2SO_4 \rightarrow Al_2(SO_4)_3 + 3H_2\uparrow$
(d) Iron + Hydrochloric acid → Iron(II) chloride + Hydrogen
$Fe + 2HCl \rightarrow FeCl_2 + H_2\uparrow$

Q6. Compounds such as alcohols and glucose also contain hydrogen but are not categorised as acids. Describe an activity to prove it.

Ans. Take glucose or alcohol solution in a beaker and connect two electrodes to a battery, bulb and switch. The bulb does not glow, showing no H^+ ions / electric current. Repeat with dilute HCl — the bulb glows. Hence alcohols and glucose do not ionise to give H^+ and are not acids.

Q7. Why does distilled water not conduct electricity, whereas rain water does?

Ans. Distilled water is free of dissolved salts/ions, so it cannot conduct. Rain water dissolves acidic gases (CO_2, SO_2, etc.) forming ions (H^+, HCO_3^-) which make it a conductor.

Q8. Why does dry HCl gas not show acidic behaviour in the absence of water?

Ans. Acidic character is due to H^+ ions, which are produced only when HCl dissociates in water. Dry HCl does not ionise, so it does not show acidic behaviour.

Q9. Five solutions A, B, C, D and E show pH 4, 1, 11, 7 and 9 respectively. Which is (a) neutral, (b) strongly alkaline, (c) strongly acidic, (d) weakly acidic, (e) weakly alkaline? Arrange the pH in increasing order of hydrogen-ion concentration.

Ans. (a) D (pH 7) – neutral (b) C (pH 11) – strongly alkaline
(c) B (pH 1) – strongly acidic (d) A (pH 4) – weakly acidic
(e) E (pH 9) – weakly alkaline.
H^+ concentration increases as pH falls: C (11) < E (9) < D (7) < A (4) < B (1).

Q10. Equal lengths of magnesium ribbon are taken in test tubes A and B. HCl is added to A and acetic acid (CH_3COOH) to B; amount and concentration of both acids are the same. In which tube will fizzing be more vigorous and why?

Ans. Fizzing is more vigorous in A. HCl is a strong acid that completely ionises giving a high H^+ concentration, while acetic acid is a weak acid and is only partially ionised, so the reaction is slower.

Q11. Fresh milk has a pH of 6. How will the pH change as it turns into curd? Explain.

Ans. pH will decrease (become more acidic). Bacteria convert lactose into lactic acid, which lowers the pH and sets the curd.

Q12. A milkman adds a very small amount of baking soda to fresh milk.
(a) Why does he shift the pH from 6 to slightly alkaline?
(b) Why does this milk take a long time to set as curd?

Ans. (a) Baking soda ($NaHCO_3$) makes the milk slightly alkaline, which slows the growth of lactic-acid bacteria and prevents the milk from turning sour quickly.
(b) The alkali neutralises the lactic acid formed, so the milk takes longer to become acidic enough to set as curd.

Q13. Plaster of Paris should be stored in a moisture-proof container. Explain why.

Ans. Plaster of Paris ($CaSO_4 \cdot \frac{1}{2}H_2O$) absorbs water to form hard gypsum ($CaSO_4 \cdot 2H_2O$), losing its setting property. It must therefore be kept dry.

Q14. What is a neutralisation reaction? Give two examples.

Ans. The reaction of an acid with a base to form salt and water is called neutralisation.
e.g., $NaOH + HCl \rightarrow NaCl + H_2O$
$Mg(OH)_2 + 2HCl \rightarrow MgCl_2 + 2H_2O$

Q15. Give two important uses of washing soda and baking soda.

Ans. **Washing soda ($Na_2CO_3 \cdot 10H_2O$):** (i) used in glass, soap and paper industries; (ii) used as a cleaning agent and for removing permanent hardness of water.
Baking soda ($NaHCO_3$): (i) used in baking powder to make cakes/bread fluffy; (ii) used in antacids and in soda-acid fire extinguishers.

CHAPTER 3 Metals and Non-metals

Section A — Multiple Choice Questions

Choose the correct option for the following questions.

Q1. Which of the following pairs will give displacement reactions?

(a) NaCl solution and copper metal
(b) $MgCl_2$ solution and aluminium metal
(c) $FeSO_4$ solution and silver metal
(d) $AgNO_3$ solution and copper metal [Correct]

Ans. Copper is more reactive than silver, so it displaces silver from $AgNO_3$: $Cu + 2AgNO_3 \rightarrow Cu(NO_3)_2 + 2Ag$. In the other pairs the metal is less reactive than the salt's metal.

Q2. Which method is suitable for preventing an iron frying pan from rusting?

(a) Applying grease
(b) Applying paint
(c) Applying a coating of zinc [Correct]
(d) All of the above

Ans. Galvanisation (zinc coating) is best for a frying pan; grease and paint would burn off during cooking.

Q3. An element reacts with oxygen to give a high-melting compound that is also soluble in water. The element is likely to be

(a) calcium [Correct]
(b) carbon
(c) silicon
(d) iron

Ans. Calcium forms CaO (high melting point) which dissolves in water to form $Ca(OH)_2$.

Q4. Food cans are coated with tin and not zinc because

(a) zinc is costlier than tin
(b) zinc has a higher melting point than tin
(c) zinc is more reactive than tin [Correct]
(d) zinc is less reactive than tin

Ans. Zinc is more reactive than tin; it could react with the food (especially acidic) and make it unfit. Tin is less reactive and safer.

Section B — Short / Numerical Questions

Q5. You are given a hammer, a battery, a bulb, wires and a switch.
(a) How could you use them to distinguish samples of metals and non-metals?
(b) Assess the usefulness of these tests.

Ans. (a) **Hammer test:** strike each sample — metals are malleable (flatten, do not break); non-metals are brittle (break). **Electrical test:** connect the sample in the circuit with battery, bulb and wires — metals make the bulb glow (conductors), non-metals do not (except graphite).
(b) These tests are simple and useful, but not universal — e.g. graphite (a non-metal) conducts electricity, and mercury (a metal) is liquid/not malleable. So results must be combined with other properties.

Q6. What are amphoteric oxides? Give two examples.

Ans. Oxides that react with both acids and bases to form salt and water are amphoteric oxides. Examples: Aluminium oxide (Al_2O_3) and Zinc oxide (ZnO).

Q7. Name two metals which will displace hydrogen from dilute acids, and two which will not.

Ans. Displace hydrogen: Zinc and iron (or magnesium, sodium).
Do not displace hydrogen: Copper and silver (or gold) — they lie below hydrogen in the reactivity series.

Q8. In the electrolytic refining of a metal M, what would you take as the anode, cathode and electrolyte?

Ans. Anode – impure metal M; Cathode – a thin strip of pure metal M; Electrolyte – a solution of a suitable salt of metal M (e.g. aqueous copper sulphate for refining copper).

Q9. Pratyush heated sulphur powder and collected the gas evolved.
(a) What will be the action of the gas on (i) dry litmus paper? (ii) moist litmus paper?
(b) Write a balanced chemical equation for the reaction.

Ans. (a) (i) No effect on dry litmus paper. (ii) Blue moist litmus turns red, because the gas forms sulphurous acid in water.
(b) $S + O_2 \rightarrow SO_2$; $SO_2 + H_2O \rightarrow H_2SO_3$ (sulphurous acid).

Q10. State two ways to prevent rusting of iron.

Ans. (i) Painting, oiling or greasing the surface; (ii) galvanisation — coating with a layer of zinc. (Chrome plating, alloying to stainless steel are other methods.)

Q11. What type of oxides are formed when non-metals combine with oxygen?

Ans. Non-metals form either acidic oxides (e.g. CO_2, SO_2) or neutral oxides (e.g. CO, H_2O, NO).

Q12. Give reasons:
(a) Platinum, gold and silver are used to make jewellery.
(b) Sodium, potassium and lithium are stored under oil.
(c) Aluminium, a highly reactive metal, is used to make cooking utensils.
(d) Carbonate and sulphide ores are usually converted into oxides during extraction.

Ans. (a) They are lustrous, highly malleable/ductile, least reactive and do not corrode, so they keep their shine.
(b) They react vigorously with oxygen and moisture in air (and can catch fire); oil cuts off air and water.
(c) Aluminium forms a protective, non-reactive layer of Al_2O_3 on its surface and is a good conductor of heat.
(d) It is easier to extract a metal from its oxide by reduction than directly from carbonate or sulphide ores.

Q13. Tarnished copper vessels are cleaned with lemon or tamarind juice. Explain why these sour substances are effective.

Ans. Tarnish is a layer of basic copper carbonate/CuO. The acids in lemon/tamarind (citric/tartaric acid) react with and dissolve this layer, restoring the shiny copper surface.

Q14. Differentiate between metal and non-metal on the basis of their chemical properties.

Ans. Metals form basic oxides, lose electrons to form positive ions, displace H from dilute acids, and form ionic chlorides. Non-metals form acidic/neutral oxides, gain electrons to form negative ions, do not displace H from dilute acids, and form covalent chlorides/compounds.

Q15. A man posing as a goldsmith dipped dull gold bangles in a solution; they sparkled like new but lost weight drastically. What was the nature of the solution?

Ans. The solution was **aqua regia** – a mixture of concentrated HCl and concentrated HNO_3 in 3:1 ratio. It dissolves gold, so an outer layer of the bangles dissolved, reducing their weight while leaving a clean, shiny surface.

Q16. Give reasons why copper is used to make hot water tanks and not steel (an alloy of iron).

Ans. Copper does not react with hot water or steam, whereas iron (steel) reacts with steam to form iron oxide and hydrogen, leading to rusting. Copper is also a better conductor of heat and resists corrosion, making it more durable.

CHAPTER 9 Light – Reflection and Refraction

Section A — Multiple Choice Questions

Choose the correct option for the following questions.

Q1. Which one of the following materials cannot be used to make a lens?

(a) Water
(b) Glass
(c) Plastic
(d) Clay [Correct]

> **Ans.** A lens must be transparent; clay is opaque.

Q2. The image formed by a concave mirror is virtual, erect and larger than the object. The object must be placed

(a) between F and C
(b) at C
(c) beyond C
(d) between the pole and the principal focus [Correct]

> **Ans.** When the object is between pole (P) and focus (F), a concave mirror forms a virtual, erect and magnified image behind it.

Q3. Where should an object be placed in front of a convex lens to get a real image of the same size as the object?

(a) At the principal focus
(b) At twice the focal length [Correct]
(c) At infinity
(d) Between optical centre and principal focus

> **Ans.** At 2F (u = 2f) a convex lens forms a real, inverted, same-sized image at 2F on the other side.

Q4. A spherical mirror and a thin spherical lens each have focal length –15 cm. They are likely to be

(a) both concave [Correct]
(b) both convex
(c) mirror concave and lens convex
(d) mirror convex and lens concave

> **Ans.** By sign convention a concave mirror and a concave (diverging) lens both have negative focal length.

Q5. No matter how far you stand from a mirror your image appears erect. The mirror is likely to be

(a) only plane
(b) only concave
(c) only convex
(d) either plane or convex **[Correct]**

> **Ans.** Plane mirrors always form erect images; convex mirrors also always form erect (virtual, diminished) images. A concave mirror gives an erect image only when the object is between P and F.

Q6. Which lens would you prefer to read small letters in a dictionary?

(a) Convex lens of f = 50 cm
(b) Concave lens of f = 50 cm
(c) Convex lens of f = 5 cm **[Correct]**
(d) Concave lens of f = 5 cm

> **Ans.** A convex lens acts as a magnifying glass; shorter focal length (5 cm) gives greater magnification when held close to the page.

Section B — Short / Numerical Questions

Q7. We wish to obtain an erect image of an object using a concave mirror of focal length 15 cm. What should be the range of the object distance? What is the nature of the image? Is it larger or smaller? Draw a ray diagram.

> **Ans.** Object must be placed between the pole and focus, i.e. $0 < u < 15$ cm. The image is virtual, erect and magnified, formed behind the mirror. (Ray diagram: object between P and F; reflected rays appear to diverge from a point behind the mirror giving an enlarged erect image.)

Q8. Name the type of mirror used in (a) headlights of a car, (b) side/rear-view mirror of a vehicle, (c) solar furnace. Give reasons.

> **Ans.** (a) **Concave mirror** – when the bulb is at its focus, it produces a powerful parallel beam of light.
> (b) **Convex mirror** – it gives an erect, diminished image and a wider field of view.
> (c) **Concave mirror** – it converges parallel sun rays to its focus, producing high heat.

Q9. One-half of a convex lens is covered with black paper. Will this lens produce a complete image of the object? Verify and explain.

> **Ans.** Yes, a complete image is formed, but it is dimmer. Light rays from each point of the object pass through the uncovered half and converge to form the complete image; fewer rays reach it, so brightness is reduced.

Q10. An object 5 cm long is held 25 cm away from a converging lens of focal length 10 cm. Draw the ray diagram and find the position, size and nature of the image.

> **Ans.** Given $h = 5$ cm, $u = -25$ cm, $f = +10$ cm.
> Using $1/v - 1/u = 1/f$: $1/v = 1/10 - 1/25 = 3/50 \rightarrow$ **$v = +16.67$ cm** (on the other side).
> $m = v/u = 16.67/(-25) = -0.67$; $h' = m \times h = -3.33$ cm.
> Image is **real, inverted and diminished**, 16.67 cm behind the lens and 3.33 cm high.

Q11. A concave lens of focal length 15 cm forms an image 10 cm from the lens. How far is the object placed? Draw the ray diagram.

Ans. Concave lens: f = –15 cm, virtual image v = –10 cm.
1/v – 1/u = 1/f → –1/10 – 1/u = –1/15 → 1/u = –1/30 → **u = –30 cm**.
Object is 30 cm in front of the lens; image is virtual, erect and diminished.

Q12. An object is placed 10 cm from a convex mirror of focal length 15 cm. Find the position and nature of the image.

Ans. u = –10 cm, f = +15 cm. 1/v + 1/u = 1/f → 1/v = 1/15 + 1/10 = 1/6 → **v = +6 cm**.
Image is 6 cm behind the mirror, virtual, erect and diminished.

Q13. The magnification produced by a plane mirror is +1. What does this mean?

Ans. m = +1 means the image is the same size as the object (|m| = 1) and it is virtual and erect (positive sign).

Q14. An object 5.0 cm long is placed 20 cm in front of a convex mirror of radius of curvature 30 cm. Find the position, nature and size of the image.

Ans. u = –20 cm, f = R/2 = +15 cm. 1/v = 1/15 + 1/20 = 7/60 → **v = +8.57 cm**.
m = –v/u = +0.43; h′ = 0.43 × 5 = **2.14 cm**.
Image is 8.57 cm behind the mirror, virtual, erect and diminished, 2.14 cm high.

Q15. An object 7.0 cm is placed 27 cm in front of a concave mirror of focal length 18 cm. At what distance should a screen be placed for a sharp image? Find its size and nature.

Ans. u = –27 cm, f = –18 cm. 1/v = 1/f – 1/u = –1/18 + 1/27 = –1/54 → **v = –54 cm**.
m = –v/u = –2; h′ = –2 × 7 = **–14 cm**.
Screen 54 cm in front of the mirror; image is real, inverted, magnified, 14 cm high.

Q16. Find the focal length of a lens of power –2.0 D. What type of lens is this?

Ans. P = 1/f (in m) → f = 1/(–2.0) = –0.5 m = **–50 cm**. Negative focal length means a **concave (diverging) lens**.

Q17. A doctor prescribes a corrective lens of power +1.5 D. Find its focal length. Is it diverging or converging?

Ans. f = 1/P = 1/1.5 = +0.667 m = **+66.7 cm**. Positive power means a **convex (converging) lens**.

CHAPTER 10 The Human Eye and the Colourful World

Section A — Multiple Choice Questions

Choose the correct option for the following questions.

Q1. The human eye can focus objects at different distances by adjusting the focal length of the eye lens. This is due to

(a) presbyopia
(b) accommodation [Correct]
(c) near-sightedness
(d) far-sightedness

Ans. The ability of the eye lens to adjust its focal length to see near and far objects clearly is called accommodation.

Q2. The human eye forms the image of an object at its

(a) cornea
(b) iris
(c) pupil
(d) retina [Correct]

Ans. The real, inverted image is formed on the retina.

Q3. The least distance of distinct vision for a young adult with normal vision is about

(a) 25 m
(b) 2.5 cm
(c) 25 cm [Correct]
(d) 2.5 m

Ans. The near point / least distance of distinct vision for a normal eye is 25 cm.

Q4. The change in focal length of an eye lens is caused by the action of the

(a) pupil
(b) retina
(c) ciliary muscles [Correct]
(d) iris

Ans. Ciliary muscles change the curvature (and hence focal length) of the eye lens.

Section B — Short / Numerical Questions

Q5. A person needs a lens of power –5.5 D for distant vision and +1.5 D for near vision. What is the focal length of the lens required for (i) distant vision, (ii) near vision?

Ans. (i) f = 1/P = 1/(–5.5) = –0.182 m = **–18.2 cm** (concave lens).
(ii) f = 1/(+1.5) = +0.667 m = **+66.7 cm** (convex lens).

Q6. The far point of a myopic person is 80 cm in front of the eye. What is the nature and power of the lens required?

Ans. A concave lens is needed so that an object at infinity forms a virtual image at 80 cm.
u = –∞, v = –80 cm; 1/f = 1/v – 1/u = –1/80 → f = –80 cm = –0.8 m.
P = 1/f = –1.25 D; a concave (diverging) lens.

Q7. Make a diagram to show how hypermetropia is corrected. The near point of a hypermetropic eye is 1 m. What is the power of the lens required? (Near point of normal eye = 25 cm.)

Ans. A convex lens converges light so that an object at 25 cm forms a virtual image at the eye's near point (100 cm).
u = –25 cm, v = –100 cm; 1/f = 1/v – 1/u = –1/100 + 1/25 = 3/100 cm^{-1}.
f = 33.3 cm = 0.333 m; **P = +3.0 D** (convex lens).

Q8. Why is a normal eye not able to see clearly objects placed closer than 25 cm?

Ans. At less than 25 cm the ciliary muscles cannot contract enough to make the eye lens sufficiently thick/converging, so light is focused behind the retina and the image appears blurred. 25 cm is the least distance of distinct vision.

Q9. What happens to the image distance in the eye when we increase the object distance?

Ans. The image distance in the eye remains unchanged — the image is always formed on the retina. Instead, the focal length of the eye lens is increased (made thinner) by the ciliary muscles to focus the distant object.

Q10. Why do stars twinkle?

Ans. Starlight passes through layers of air of changing optical density/refractive index. It is refracted continuously, so the apparent position and brightness fluctuate. This atmospheric refraction causes the twinkling of stars.

Q11. Explain why the planets do not twinkle.

Ans. Planets are much closer and appear as extended collections of point sources. The variations in light from different points average out, so the total brightness remains steady and planets do not appear to twinkle.

Q12. Why does the sky appear dark instead of blue to an astronaut?

Ans. There is no atmosphere in space to scatter sunlight. With no scattering of blue light, no light reaches the astronaut's eyes from the sky, so it appears dark (black).